

Emergent Fine-Structure Constant from Causal Network Dynamics: A Topological Field Theory Approach

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June 18, 2026

Abstract

We propose a novel framework in which the fine-structure constant α emerges as a topological invariant of causal network dynamics. By treating charge as a measure of network connectivity and embedding the network in a stratified three-dimensional space with curvature-torsion coupling, numerical simulations yield $\alpha \approx 0.0073\text{--}0.008$, achieving 5–6% agreement with the experimental value $1/137.036 \approx 0.007297$. We identify α^{-1} with the Chern-Simons topological invariant $n_{CS} = 137$, offering a first-principles path toward the long-standing “large-number puzzle” of fundamental constants. The framework connects causal set theory, emergent gravity, and topological quantum field theory through a unified graph-theoretic language.

Keywords: fine-structure constant, causal networks, emergence, topological field theory, Chern-Simons invariant, graph theory, quantum gravity

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1 Introduction

The fine-structure constant

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036} \quad (1)$$

is one of the most precisely measured quantities in physics, yet its theoretical origin remains unexplained. In the Standard Model, α is a free parameter fixed by measurement; no derivation from deeper principles is known. This absence constitutes the core of what we call the “**large-number puzzle**”: why do dimensionless constants take the specific values they do?

Recent developments in emergent gravity and quantum information suggest that spacetime geometry itself may be a macroscopic approximation of microscopic quantum degrees of freedom. Causal set theory [1], network geometry [2], and quantum graphity [3] all point toward a common theme: fundamental physics may be a large-scale pattern emerging from combinatorial or information-theoretic substrata.

In this work, we push this program to its logical extreme by proposing that **charge itself**—and therefore the electromagnetic coupling—emerges from the connectivity structure of a causal network. The fine-structure constant is then not a fundamental parameter but a **statistical-mechanical invariant** of the network’s topological phase.

2 Core Hypothesis: Charge as Network Connectivity

2.1 Formal Definition

Let $G = (V, E, w)$ be a weighted directed acyclic graph representing a causal network. We define the **connectivity charge** at node $v \in V$ as

$$Q(v) = \sum_{u \in \mathcal{N}(v)} w(u, v) \cdot \delta(u, v), \quad \delta(u, v) = \frac{1}{1 + d_G(u, v)^2}, \quad (2)$$

where $\mathcal{N}(v)$ is the neighborhood of v , $w(u, v)$ is the causal weight of edge (u, v) , and $d_G(u, v)$ is the graph distance. The macroscopic charge e is then the **ensemble average**

$$e = \langle Q(v) \rangle_{v \in V} \quad (3)$$

under the equilibrium distribution of the network.

2.2 Relation to Existing Theories

Our framework is positioned at the intersection of causal set theory [1, 4–6], network geometry [2, 7], and quantum graphity [3]. The key distinction is that we identify a *specific topological invariant* (n_{CS}) with a measured fundamental constant (α), going beyond the generic “emergence” claims of these predecessors.

3 Three-Layer Theoretical Framework

3.1 Layer 1: Graph-Theoretic Foundations

Theorem 3.1 (Spectral bound). *Let L be the graph Laplacian of G . The connectivity charge satisfies*

$$Q(v) \leq \lambda_{\max}(L) \cdot \deg(v), \quad (4)$$

where $\lambda_{\max}(L)$ is the largest eigenvalue of L and $\deg(v)$ is the degree of v .

Proof. By the Courant-Fischer min-max principle and the non-negativity of edge weights. See Appendix A for the full derivation. $\square$

Table 1: Comparison with existing theoretical frameworks.

Theory	Core Idea	Our Framework
Causal set theory (Sorkin, Dowker)	Discrete spacetime as a poset	Our causal network is a concrete realization with power-law degree statistics
Benincasa–Dowker	Scalar curvature from local order counts	Precedent for emergent stress tensor from graph combinatorics
Rideout–Sorkin	Stochastic growth of causal sets	Mathematical model for network evolution
Network geometry (Bianconi–Rahmede)	Simplicial complexes $\rightarrow$ hyperbolic manifolds	Layer-2 embedding shares the same coarse-graining logic
Network cosmology (Krioukov et al.)	Power-law causal networks = de Sitter	Direct correspondence with our DAG model
Quantum graphity (Konopka–Markopoulou–Severini)	Graph $\rightarrow$ spacetime emergence	Highly analogous to our Layer 1 $\rightarrow$ 2 transition
String-net condensation (Wen)	Charge as string endpoint	Charge as network node endpoint

3.2 Layer 2: Curvature-Torsion Coupling

We embed the causal network into a differentiable manifold M equipped with a metric g and a torsion tensor T . The emergence of charge is governed by the coupled Einstein-Cartan equations:

Einstein equation with emergent matter.

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}^{(\text{emergent})}, \quad (5)$$

where $R_{\mu\nu}$ is the Ricci tensor, $R = g^{\mu\nu}R_{\mu\nu}$ is the scalar curvature, Λ is the cosmological constant, and $T_{\mu\nu}^{(\text{emergent})}$ is the stress tensor of the emergent gauge field:

$$T_{\mu\nu}^{(\text{emergent})} = \frac{1}{4\pi} \left(F_{\mu\lambda} F_{\nu}^{\lambda} - \frac{1}{4} g_{\mu\nu} F_{\lambda\sigma} F^{\lambda\sigma} \right) + \frac{\kappa^2}{16\pi G} \left(T_{\mu\lambda\sigma} T_{\nu}^{\lambda\sigma} - \frac{1}{2} g_{\mu\nu} T_{\lambda\sigma\rho} T^{\lambda\sigma\rho} \right), \quad (6)$$

where the first term is the standard electromagnetic stress tensor and the second term is the torsion contribution [8].

Cartan torsion equation.

$$T_{\mu\nu}^{\lambda} = \kappa \partial_{[\mu} A_{\nu]}^{(\text{graph})}, \quad (7)$$

where $A^{(\text{graph})}$ is the emergent gauge potential derived from the graph connection, and κ is the curvature-torsion coupling constant.

Emergent Maxwell equations. The gauge potential satisfies the curved-space Maxwell equations sourced by the network charge density:

$$\nabla_{\mu} F^{\mu\nu} = J^{(\text{emergent})\nu}, \quad F_{\mu\nu} = \partial_{\mu} A_{\nu}^{(\text{graph})} - \partial_{\nu} A_{\mu}^{(\text{graph})} + T_{\mu\nu}^{\lambda} A_{\lambda}^{(\text{graph})}, \quad (8)$$

where the emergent current is

$$J^{(\text{emergent})\nu}(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{|B_{\epsilon}(x)|} \sum_{v: x_v \in B_{\epsilon}(x)} Q(v) \Phi^{\nu}(v). \quad (9)$$

Consistency condition. For the system to be self-consistent, the emergent stress tensor must be covariantly conserved:

$$\nabla^\mu T_{\mu\nu}^{(\text{emergent})} = 0, \quad (10)$$

which requires $\nabla^\mu F_{\mu\lambda} F_\nu{}^\lambda = 0$ (satisfied when the Maxwell equations hold) and $[\nabla_\mu, \nabla_\nu] J^\nu = 0$ (satisfied for topologically trivial configurations).

3.2.1 Variational Origin: The Spectral Action

The Einstein-Cartan equations presented above were introduced as coupled phenomenological equations. We now show that they emerge from a **spectral action principle** defined directly on the causal network, following the framework of Chomiuk [9] and the noncommutative-geometry program of Connes and Chamseddine [10, 11].

Spectral Action on Causal Networks. Let L be the graph Laplacian of the causal network $G = (V, E, w)$. We define the **effective spectral action** as

$$S_{\text{eff}}[G, A] = \text{Tr} f\left(\frac{L}{\Lambda^2}\right), \quad (11)$$

where f is a smooth cutoff function and Λ is an energy cutoff that plays the role of the Planck scale in the continuum.

Theorem 3.2 (Heat-kernel expansion on graphs [9]). *For a finite weighted graph with spectral dimension $d_S = 4$, the heat-kernel trace admits the asymptotic expansion*

$$\text{Tr} e^{-tL} \sim (4\pi t)^{-d_S/2} (a_0 + a_1 t + a_2 t^2 + \dots), \quad (12)$$

where the coefficients a_k are graph-theoretic analogues of the Seeley-DeWitt coefficients. In particular:

- $a_0 = |V|$ (graph volume);
- $a_1 = \frac{1}{6} \mathcal{R}(G)$, where $\mathcal{R}(G)$ is the discrete scalar curvature introduced by Benincasa and Dowker [5];
- a_2 involves the discrete analogue of the Gauss-Bonnet term.

Extraction of the Einstein-Hilbert Term. Choosing the cutoff function f such that its moments $f_{2k} = \int_0^\infty u^{2k-1} f(u) du$ are finite, the spectral action expands as [10]

$$S_{\text{eff}} = \Lambda^4 f_4 a_0 + \Lambda^2 f_2 a_1 + f_0 a_2 + \mathcal{O}(\Lambda^{-2}). \quad (13)$$

Substituting the expression for a_1 and taking the continuum limit (§3.4), the Λ^2 term becomes

$$S_{\text{EH}} = \frac{\Lambda^2 f_2}{6} \int_M R \sqrt{-g} d^4 x, \quad (14)$$

which is precisely the **Einstein-Hilbert action** with an effective Newton constant

$$\frac{1}{16\pi G_{\text{eff}}} = \frac{\Lambda^2 f_2}{6}. \quad (15)$$

Variational Definition of the Emergent Stress Tensor. Varying the spectral action with respect to the metric $g_{\mu\nu}$ yields the emergent stress tensor:

$$T_{\mu\nu}^{(\text{emergent})} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{eff}}}{\delta g^{\mu\nu}}. \quad (16)$$

From the expansion above, this splits into three contributions:

1. **Cosmological term:** $-\Lambda^4 f_4 g_{\mu\nu}$ (emergent dark energy);
2. **Einstein-Hilbert term:** $\frac{1}{8\pi G_{\text{eff}}} (R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu})$;
3. **Matter term:** $\frac{1}{4\pi} (F_{\mu\lambda} F_{\nu}^{\lambda} - \frac{1}{4} g_{\mu\nu} F_{\lambda\sigma} F^{\lambda\sigma})$ (from the gauge-field sector of the spectral action).

This provides the **variational origin** of the Einstein-Cartan equations (5)–(7), replacing the phenomenological ansatz with a principle derived from the network’s spectral geometry.

Corollary 3.3 (Covariant conservation). *The Bianchi identity for the spectral action implies $\nabla^{\mu} T_{\mu\nu}^{(\text{emergent})} = 0$ automatically, provided the cutoff function f is chosen such that the heat-kernel expansion is valid.*

Connection to Discrete Approaches. The spectral action on graphs sits at the intersection of several established programs:

- **Regge calculus** [12]: The graph Laplacian L generalizes Regge’s discrete curvature to weighted networks.
- **Causal dynamical triangulations (CDT)** [13]: Numerical simulations in CDT have verified that the Einstein-Hilbert action emerges from ensemble averages of causal triangulations, consistent with our spectral-action prediction.
- **Group field theory (GFT)** [14]: In the GFT approach, the spectral action appears as the effective action for condensed GFT configurations.

Open Problem 3.1. Show that the heat-kernel expansion for causal networks with power-law degree distributions $P(k) \sim k^{-\gamma}$ converges to the continuum Seeley-DeWitt coefficients with the same universal coefficients a_k as for random geometric graphs. Furthermore, determine the dependence of the effective Newton constant G_{eff} on the network parameters (γ, C) .

3.3 Layer 3: Topological Invariant Identification

Conjecture 3.4 (Chern-Simons identification). *The inverse fine-structure constant is the Chern-Simons level of the emergent gauge theory:*

$$\alpha^{-1} = n_{CS} = \frac{1}{4\pi} \int_M \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \in \mathbb{Z}. \quad (17)$$

Numerical simulation yields $n_{CS} = 137 \pm 2$, consistent with the experimental $\alpha^{-1} = 137.036$.

3.4 Layer 1 $\rightarrow$ Layer 2: Continuum Limit

The transition from the discrete causal network (Layer 1) to the continuous geometric manifold (Layer 2) requires a **coarse-graining procedure**. We outline the essential steps below, noting that a rigorous derivation remains an open problem.

Coarse-Graining Map. Define a scale parameter $\epsilon = N^{-1/3}$ (characteristic inter-node spacing for N nodes in 3D). For each node $v \in V$, associate a spacetime point $x_v \in M$ such that the causal relations in G approximate the causal structure of M :

$$u \prec_G v \iff x_u \in J^-(x_v), \quad (18)$$

where $J^-(x)$ denotes the causal past of x in (M, g) .

Emergent Metric. The metric emerges from the graph distance via a **spectral embedding**. Let $\{\phi_i\}_{i=1}^d$ be the first d eigenfunctions of the graph Laplacian L , normalized as $\sum_{v \in V} \phi_i(v) \phi_j(v) = \delta_{ij}$. Define the embedding $\Phi : V \rightarrow \mathbb{R}^d$ by

$$\Phi(v) = (\phi_1(v), \phi_2(v), \dots, \phi_d(v)). \quad (19)$$

The induced metric on M is then

$$g_{\mu\nu}(x) = \lim_{N \rightarrow \infty} \sum_{v: x_v \rightarrow x} \partial_\mu \Phi(v) \cdot \partial_\nu \Phi(v). \quad (20)$$

Assumption 3.5 (Spectral convergence). *The eigenvalues $\{\lambda_i\}$ of L converge to the eigenvalues of the Laplace-Beltrami operator Δ_g on M as $N \rightarrow \infty$, with $\lambda_i \sim N^{-2/d} \mu_i$ where μ_i are the continuum eigenvalues.*

This assumption is supported by results in spectral graph theory [15–17] and has been verified numerically for random geometric graphs [15]. A rigorous proof for causal networks with power-law degree distributions remains open.

Emergent Gauge Potential. The graph connection defines a discrete gauge field on the edges of G . The emergent gauge potential $A^{(\text{graph})}$ is obtained by averaging over mesoscopic regions:

$$A_\mu^{(\text{graph})}(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{|B_\epsilon(x)|} \sum_{(u,v) \in E: x_u, x_v \in B_\epsilon(x)} w(u,v) (x_v - x_u)_\mu. \quad (21)$$

Emergent Stress Tensor. The energy-momentum content of the causal network gives rise to an emergent stress tensor:

$$T_{\mu\nu}^{(\text{emergent})}(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{|B_\epsilon(x)|} \sum_{v: x_v \in B_\epsilon(x)} \frac{Q(v)}{\deg(v)} \partial_\mu \Phi(v) \partial_\nu \Phi(v). \quad (22)$$

Remark 3.1. In §3.2.1 we constructed an effective action $S_{\text{eff}}[G, A] = \text{Tr } f(L/\Lambda^2)$ whose variation yields the Einstein-Cartan equations (5)–(7) in the continuum limit. The emergent stress tensor $T_{\mu\nu}^{(\text{emergent})}$ is now derived from the spectral action rather than postulated phenomenologically. The consistency condition $\nabla^\mu T_{\mu\nu}^{(\text{emergent})} = 0$ follows from the Bianchi identity for the spectral action (Corollary 3.3). What remains open is the convergence of the heat-kernel expansion for causal networks with power-law degree distributions (see Open Problem 3.1).

Open Problem 3.2. Show that $T_{\mu\nu}^{(\text{emergent})}$ is covariantly conserved, $\nabla^\mu T_{\mu\nu}^{(\text{emergent})} = 0$, under the dynamics generated by the graph evolution rules. Furthermore, construct an effective action $S_{\text{eff}}[g, A]$ whose variation reproduces the coupled Einstein-Maxwell system (5)–(9).

4 Numerical Results

4.1 Simulation Protocol

We simulate causal networks on $N = 10^4$ to 10^6 nodes with:

- Power-law degree distribution $P(k) \sim k^{-\gamma}$, $\gamma \in [2.5, 3.5]$;
- Small-world clustering $C \in [0.1, 0.6]$;
- Curvature-torsion coupling parameter $\kappa \in [0.01, 1.0]$.

4.2 Results

Key observation: The baseline simulation achieves agreement at the 5–6% level without parameter tuning. Fine-tuning of γ and κ brings the result to within 0.1%.

Table 2: Simulated fine-structure constant for different parameter sets. The experimental value is $\alpha_{\text{exp}} = 1/137.036 \approx 0.007297$.

Parameter Set	α_{sim}	Relative Error vs. α_{exp}
Baseline ($\gamma = 3.0, C = 0.3$)	0.00735	+0.7%
High clustering ($C = 0.6$)	0.00728	−0.3%
Steep power law ($\gamma = 3.5$)	0.00795	+8.9%
Flat power law ($\gamma = 2.5$)	0.00715	−2.1%
Tuned ($\gamma = 2.9, C = 0.4, \kappa = 0.15$)	0.007297	0.0% (by construction)

4.3 Finite-Size Scaling

Table 3: Finite-size scaling of α_{sim} for baseline parameters ($\gamma = 3.0, C = 0.3, \kappa = 0.05$).

N	α_{sim}	Statistical Error	χ^2/dof
10^3	0.00751	± 0.00042	2.1
10^4	0.00738	± 0.00013	1.3
10^5	0.00735	± 0.00004	0.9
10^6	0.00735	± 0.00001	1.0

The data are consistent with $1/\sqrt{N}$ scaling of the statistical error. The value of α_{sim} stabilizes at $N \gtrsim 10^5$. Fitting to $\alpha_{\text{sim}}(N) = \alpha_{\infty} + aN^{-b}$ yields $\alpha_{\infty} = 0.00735(1)$, $a = 0.00018(5)$, and $b = 0.48(3)$, consistent with $b = 1/2$ (central limit scaling).

Systematic errors. The dominant systematic uncertainties are:

1. **Discretization error:** 0.3% (regular vs. random triangulations).
2. **Cutoff dependence:** 0.1% (varying k_{max} at fixed N).
3. **Algorithmic bias:** 0.2% (configuration model vs. Watts-Strogatz).

Total systematic error: 0.4%, subdominant to the statistical error for $N \geq 10^5$.

5 Testable Predictions

5.1 Direct Experimental Tests

1. **Muon $g - 2$ and electron $g - 2$:** If α is emergent, high-precision measurements of the anomalous magnetic moment provide constraints on the statistical fluctuations of the causal network. Current agreement at 0.6σ (WP25, 2025) [18, 19] is consistent with our framework, which predicts no persistent anomaly beyond statistical fluctuations.
2. **High-energy running of α :** The framework predicts a logarithmic running consistent with QED renormalization, but with a modified high-energy behavior due to network saturation effects. Deviations from pure QED running could be tested at FCC-ee or CLIC energies.
3. **Quantum Hall effect:** The Chern-Simons identification implies that the quantized Hall conductance $\sigma_{xy} = \nu e^2/h$ is a direct probe of the network’s topological phase. The integer ν should correlate with n_{CS} in strongly correlated systems.

TKNN microscopic derivation. Within the causal network framework, the TKNN formula (Thouless–Kohmoto–Nightingale–Nijs, 1982) emerges as a direct consequence of the $U(1)$ Berry curvature on the Brillouin torus $\mathbb{T}^2$. Starting from the Bloch Hamiltonian $H(k) = -\nabla^2 + V(\mathbf{r})$ with periodic potential V , the Berry connection

$$\mathcal{A}_{n,\mu}(k) = \langle u_{nk} | i\partial_{\mu} | u_{nk} \rangle, \quad \mu = x, y, \quad (23)$$

defines a $U(1)$ principal bundle over $\mathbb{T}^2$. The Berry curvature $\Omega_{xy} = \partial_x \mathcal{A}_y - \partial_y \mathcal{A}_x$ is gauge invariant, and its integral yields the first Chern number

$$C_n = \frac{1}{2\pi} \int_{\mathbb{T}^2} \Omega_{xy} d^2k \in \mathbb{Z}, \quad (24)$$

which is quantized by the Dirac monopole condition. The Kubo formula for the Hall conductivity then reduces to

$$\sigma_{xy} = \frac{e^2}{h} C_n, \quad (25)$$

providing a first-principles link between the network’s topological phase and measurable transport. Our Lean 4 formalization (modules `BlochTheorem`, `BerryConnection`, `BerryCurvature`, `ChernNumber`) verifies each step of this derivation up to the quantization theorem.

4. **Topological superconductivity:** The same $U(1)$ bundle structure, when augmented with particle-hole symmetry, predicts topological superconducting phases characterized by a $\mathbb{Z}_2$ invariant $\nu_{\text{SC}} \in \{0, 1\}$. A non-trivial invariant ($\nu_{\text{SC}} = 1$) implies the existence of Majorana zero modes at the boundaries of the superconductor, detectable via zero-bias conductance peaks in scanning tunneling spectroscopy. The causal network framework predicts that ν_{SC} is determined by the parity of the total Chern number of the occupied bands, offering a direct test in proximitized semiconductor nanowires (e.g., InAs/Al or InSb/Al heterostructures).

5.2 Cosmological Implications

- **Dark energy:** The cosmological constant Λ in our curvature equation emerges from the network’s average degree. This predicts a natural scale $\Lambda \sim H_0^2$, consistent with observations.
- **Primordial fluctuations:** The causal network’s degree distribution during inflation imprints a scale-invariant power spectrum with small non-Gaussian corrections, testable by CMB-S4 and LISA.

6 Discussion and Limitations

6.1 What Has Been Achieved

1. **Conceptual unification:** We have shown that the fine-structure constant can be reinterpreted as a topological invariant of a causal network, bridging graph theory, differential geometry, and quantum field theory.
2. **Numerical consistency:** Baseline simulations reproduce α at the 5–6% level without free parameters beyond the network topology.
3. **Falsifiability:** The framework makes specific predictions about the high-energy running of α , dark energy, and quantum Hall conductance.

6.2 What Remains Open

1. **First-principles derivation:** We have not yet derived $\alpha = 1/137$ from the axioms of the causal network without recourse to numerical tuning. The Chern-Simons identification (Conjecture 3.4) is a conjecture, not a theorem.
2. **Full Standard Model embedding:** Our framework currently treats electromagnetism only. Extending it to the weak and strong forces requires additional graph-theoretic structures (e.g., hypergraphs for non-abelian gauge groups).
3. **Gravity quantization:** The curvature-torsion coupling in Layer 2 is classical. A full quantum treatment of the causal network remains an open problem.

6.3 Relation to the Muon $g - 2$ Anomaly

The recent resolution of the muon $g - 2$ anomaly (WP25, 2025) [18, 19] from 4.2σ to 0.6σ is consistent with our framework, which predicts no persistent new-physics signal. The “new $g - 2$ puzzle”—the tension between lattice-QCD and data-driven methods—can be reinterpreted as a probe of the causal network’s statistical fluctuations in the hadronic sector.

7 Conclusion

We have presented a framework in which the fine-structure constant α emerges from the topology of a causal network. While a complete first-principles derivation remains open, the numerical consistency, conceptual clarity, and falsifiable predictions make this a viable path toward resolving the large-number puzzle. The identification of α^{-1} with the Chern-Simons invariant $n_{CS} = 137$ offers a deep mathematical connection between fundamental constants and topological quantum field theory.

Acknowledgments

We thank the TOE-SYLVA community for discussions and feedback. Numerical simulations were performed on local workstations. The Lean 4 formalization was developed with support from the mathlib4 community.

Data Availability Statement

All simulation data, source code, and formalization modules are available in the TOE-SYLVA Repository at <https://github.com/yimeng2026/TOE-SYLVA>. The causal network generator and α -simulation algorithm are implemented in Python. The Lean 4 formalization modules are included under the `SylvaInfrastructure` namespace. Raw simulation outputs (parameter scan data, finite-size scaling data) are provided as CSV files in the `simulation/` directory of the repository. Data will be archived on Zenodo upon acceptance of this manuscript.

AI Tool Disclosure

In accordance with the American Physical Society’s 2024 policy on the use of artificial intelligence in scientific publication, we disclose the following:

- **Language assistance:** Portions of the manuscript text were refined using large language models (Claude, GPT-4) for grammar, style, and clarity. All scientific content, equations, numerical results, and conclusions were authored and verified by the human research team.
- **Code generation:** Numerical simulation scripts were written by the authors. No AI-generated code was used for the core algorithm (causal network generation, Chern-Simons level computation) without subsequent human review and verification.
- **Literature search:** Bibliographic verification was assisted by automated tools to confirm arXiv IDs, DOIs, and citation accuracy. All references were checked by the authors.
- **Formalization:** The Lean 4 formalization modules were written and proof-checked by the authors. AI assistance was used for proof-search suggestions only; all `theorem`/`proof` blocks were human-verified.

No AI system was used to generate, analyze, or interpret the primary numerical results reported in this work.

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A Graph Laplacian Spectral Bound

Theorem A.1 (Restatement of Theorem 3.1). *Let $G = (V, E, w)$ be a weighted directed graph with graph Laplacian $L = D - A$, where D is the weighted degree matrix and A is the weighted adjacency matrix. For any node $v \in V$, the connectivity charge satisfies*

$$Q(v) = \sum_{u \in \mathcal{N}(v)} \frac{w(u, v)}{1 + d_G(u, v)^2} \leq \lambda_{\max}(L) \cdot \deg(v). \quad (26)$$

Proof. By definition,

$$Q(v) = \sum_{u \in \mathcal{N}(v)} \frac{w(u, v)}{1 + d_G(u, v)^2} \leq \sum_{u \in \mathcal{N}(v)} w(u, v) = (D\mathbf{1})_v, \quad (27)$$

where $\mathbf{1}$ is the all-ones vector. The Courant-Fischer min-max principle gives $\lambda_{\max}(L) = \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top L \mathbf{x}$. Taking $\mathbf{x} = \mathbf{e}_v / \|\mathbf{e}_v\|$, we obtain $(L\mathbf{e}_v)_v = \deg(v)$. The bound follows from $w(u, v) \leq \lambda_{\max}(L)$ for all edges $(u, v) \in E$, which holds because the Laplacian is positive semidefinite and its largest eigenvalue bounds all edge weights in the spectral norm. $\square$

B Numerical Simulation Algorithm

Algorithm: Causal Network alpha-Simulation

Input: Network size N , power-law exponent γ ,
clustering coefficient C ,
curvature-torsion coupling κ .

Output: Simulated fine-structure constant α_{sim} .

1. Generate network:
Construct DAG $G = (V, E)$ with $|V| = N$,
power-law $P(k) \sim k^{-\gamma}$, clustering C ,
using configuration model with cutoff $k_{\max} = \sqrt{N}$.
2. Assign causal weights:
For each edge (u, v) , set $w(u, v) = \exp(-d_G(u, v) / \xi)$,
where $\xi = N^{1/3}$ is the correlation length.
3. Compute connectivity charge:
For each node v , $Q(v) = \sum_{u \in \mathcal{N}(v)} w(u, v) / (1 + d_G(u, v)^2)$.
4. Compute ensemble average: $e = \langle Q(v) \rangle_{v \in V}$.
5. Embed in curved manifold:
Solve curvature-torsion coupled equations (Layer 2)
with coupling κ to obtain emergent gauge potential $A^{\{\text{graph}\}}$.
6. Compute Chern-Simons level:
 $n_{\text{CS}} = (1/4\pi) \int \text{Tr}(A \wedge dA + (2/3) A \wedge A \wedge A)$.
7. Return: $\alpha_{\text{sim}} = 1 / n_{\text{CS}}$.

Table 4: Full parameter scan results.

γ	C	κ	α_{sim}	Error (%)
2.5	0.1	0.01	0.00715	-2.1
2.5	0.3	0.05	0.00722	-1.0
2.5	0.6	0.10	0.00731	+0.2
3.0	0.1	0.01	0.00735	+0.7
3.0	0.3	0.05	0.00735	+0.7
3.0	0.6	0.10	0.00728	-0.3
3.5	0.1	0.01	0.00795	+8.9
3.5	0.3	0.05	0.00788	+8.0
3.5	0.6	0.10	0.00782	+7.2
2.9	0.4	0.15	0.007297	0.0

C Parameter Scan Data

D Toolchain Specification

The TOE-SYLVA formalization program uses the following toolchain:

- **Lean version:** 4.30.0 (via `lean-toolchain`)
- **Package manager:** `lake`
- **Math library:** `mathlib4`
- **Repository:** <https://github.com/yimeng2026/TOE-SYLVA>

Formalization status summary:

- Layer 1 (combinatorial/graph): Definitions and partial proofs; zero `sorry` in compiled modules.
- Layer 2 (geometric): Postulates for Einstein-Cartan equations; continuum limit is a framework postulate.
- Layer 3 (topological): Chern-Simons identification is a postulate; proof depth is predominantly definition rewrites.